

PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

ON

WasteChain - Smart Recycling Traceability Platform

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

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Submitted by:

Pushpinder Singh – 2210990693

Narinder Singh – 2210990598

Lokesh Vaid – 2210990545

Nipun Kumar – 2210990615

Supervised By:

Dr. Monika Aggarwal



**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING AND TECHNOLOGY
CHITKARA UNIVERSITY, PUNJAB, INDIA**

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DECLARATION

We hereby certify that the work which is being presented in the project report entitled “**WasteChain**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Monika Aggarwal. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura	(Pushpinder Singh)	(Lokesh Vaid)	(Narinder Singh)	(Nipun Kumar)
Date: 15-05-2026	(2210990693)	(2210990545)	(2210990598)	(2210990615)

This is to certify that the above statement made by the candidates is correct to the best of my knowledge and belief.

(Dr. Monika Aggarwal)
Head of IOHE
Department of Computer Science and Engineering
Chitkara University Institute of Engineering and Technology,
Chitkara University, Punjab, India

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Pushpinder Singh
2210990693

Lokesh Vaid
2210990545

Narinder Singh
2210990598

Nipun Kumar
2210990615

ABSTRACT

WasteChain — Smart Recycling Traceability Platform is a web-based recyclable waste management system designed to improve transparency, efficiency, and coordination between citizens, collectors, recyclers, and administrators. Traditional waste management systems often face issues such as lack of digital tracking, inefficient waste collection, poor transparency, low citizen engagement, and difficulty in monitoring recyclable waste movement.

The proposed system provides a centralized platform where citizens can submit recyclable waste, collectors can manage pickups, and recyclers can process waste in a transparent manner. The platform also includes real-time waste tracking, AI-based waste classification, QR code generation, reward mechanisms, and role-based dashboards for efficient management.

The frontend of the system is developed using React.js with Vite, while Node.js and Express.js are used for backend services and MongoDB database management. JWT authentication is integrated to provide secure login and role-based access control, while Google Gemini AI enhances intelligent waste analysis. The system aims to improve recycling transparency, encourage citizen participation, optimize waste management operations, and simplify recyclable waste tracking through a modern digital solution.

1. INTRODUCTION

1.1 Problem Background

On paper, recycling sounds simple. But step inside—it's messy. Citizens don't have a proper system. They depend on local vendors with random pickups and unclear schedules. Sometimes waste gets collected, sometimes not. No consistency. Collectors face their own issues: they can't track requests properly, there is no structured workflow, and no transparency. Just manual coordination most of the time. Recyclers struggle too—they don't get clear visibility into incoming recyclable waste. Records are incomplete, waste movement becomes hard to track, delays happen, and confusion grows. Some systems exist, but they only solve small parts. There is no end-to-end traceability and no intelligent automation.

1.2 Objective of the Project

WasteChain addresses these challenges by providing a unified platform where citizens, collectors, recyclers, and administrators interact openly. The system enables transparent waste tracking, ensures recyclable waste is not lost, and tracks everything automatically. AI helps identify waste, QR codes improve traceability, and rewards encourage participation. The end goal is better efficiency, less confusion, and smarter recycling.

1.3 Proposed Solution / System Overview

Users log in and choose their role—citizen, collector, recycler, or admin. Citizens see a dashboard with waste submissions, rewards, and tracking history. They can submit recyclable waste, upload images, and use AI analysis. The system then generates a QR code. Collectors see nearby pickups and update collection status. Recyclers receive processed waste details. Admins monitor analytics, track users, review activity, and control approvals. The system seamlessly handles authentication, dashboards, AI analysis, QR tracking, real-time updates, analytics, and reward workflows.

1.4 Key Features or Functionality

Real-time waste tracking is at the heart of the platform. QR code traceability keeps waste movement transparent. AI waste analysis helps classify recyclable materials accurately. Dashboards show useful statistics without clutter. The reward system encourages citizen

participation. The notification module updates waste status, tracks activity, and prevents waste from getting lost in the process.

1.5 Technology / Method Used

The frontend is built using React with Vite for fast loading and a clean UI. JWT handles authentication securely. Node.js and Express manage backend services. MongoDB stores application data efficiently. Google Gemini AI powers waste image analysis. The client-server architecture keeps things structured and scalable.

1.6 Expected Outcome & Benefits

Citizens understand waste tracking better with no blind guessing. Collectors manage pickups more efficiently. Recyclers get better visibility. Admins make informed decisions. Transparency improves, manual confusion reduces, and recycling participation increases. Less delay and less stress overall as the workflow becomes much smoother.

1.7 Original Contribution Statement

WasteChain combines recycling, AI intelligence, digital traceability, and reward engagement into one unified platform. Most platforms treat these as separate concerns. WasteChain integrates them all—that is the key differentiator.

2. METHODOLOGY

2.1 Requirement Analysis

The project requirements were identified by analyzing problems in traditional waste management systems. Features like waste submission, QR-based tracking, AI waste classification, reward management, real-time notifications, analytics dashboards, and secure authentication were finalized.

2.2 System Design

The platform was designed using a client-server architecture. Separate modules were created for citizens, collectors, recyclers, administrators, waste tracking, AI analysis, rewards, and notifications.

2.3 Development

The frontend was developed using React.js and Vite, while Node.js and Express.js were used for backend development and MongoDB for database management. JWT was integrated for secure authentication and role-based access control, while Google Gemini AI was integrated for intelligent waste classification.

2.4 Testing

Functional testing, authentication testing, API testing, AI module testing, and interface testing were performed to identify and fix errors during development.

2.5 Deployment

After testing, the system was deployed and optimized for stable performance, smooth user interaction, and future scalability.

SYSTEM DESIGN

2.6 Architecture

The platform uses a basic client-server model. The frontend communicates with the backend through REST APIs. Data flows efficiently, updates happen in real time, and waste tracking works seamlessly. The AI module connects when needed, QR tracking stays integrated, and everything remains synchronized.

2.7 Modules

- Authentication — login, signup, secure user access
- Citizen Dashboard — waste submission and tracking view
- Collector Dashboard — pickup management and collection control
- Recycler Dashboard — recyclable waste processing management
- QR Tracking — real-time recyclable waste traceability
- AI Waste Analysis — intelligent waste classification support

2.8 Data Flow

Level 0: User interacts with system → processing modules → database

Level 1: Authentication → Dashboard → Waste Submission → AI Analysis → QR Tracking → Waste Collection → Recycling

2.9 Use Cases

Actors: Citizen, Collector, Recycler, Administrator

Use Cases:

- Login / Register
- Submit Waste
- AI Waste Analysis
- Track Waste
- Manage Waste Collection

2.10 ER (Entity Relationship)

Entities:

- User
- Waste Submission
- Traceability Log
- Reward System

3. TOOLS AND TECHNOLOGIES

WasteChain employs a wide range of modern tools and technologies that ensure scalability, security, transparency, and real-time waste tracking capabilities.

React.js (with Vite): To build a highly responsive, dynamic, and user-friendly front-end interface. This allows citizens, collectors, recyclers, and administrators to easily access their respective dashboards.

Node.js & Express.js: Used for backend development to handle server-side logic, API management, and smooth communication between different system modules.

MongoDB: Used for database management to securely store user data, waste submissions, traceability logs, rewards, and system records efficiently.

JWT Authentication: Provides secure user authentication and role-based access control for citizens, collectors, recyclers, and administrators.

REST APIs: Used to efficiently communicate between front-end and back-end services while ensuring secure and structured data transmission.

Google Gemini AI: Integrated for intelligent waste image analysis and classification to support accurate waste categorization.

4. IMPLEMENTATION

4.1 Technologies Used

- Frontend: React (Vite)
- Backend: Node.js + Express.js
- Database: MongoDB
- Authentication: JWT
- AI Integration: Google Gemini AI

4.2 Functionalities

- Waste submission and management
- AI-based waste classification
- QR code waste tracking
- Dashboard analytics
- Reward and gamification system
- Real-time notifications

4.3 Workflow

- User logs in
- Selects their role
- Accesses dashboard
- Submits or manages waste activities
- Tracks waste lifecycle and rewards

5. MAJOR FINDINGS / OUTCOMES / OUTPUT / RESULTS

The system improved operations clearly. Waste tracking became more transparent and manual coordination reduced significantly. Waste collection management improved, recycling visibility increased, and operations became faster with less confusion. Citizen participation improved through the reward system. There is still room for advanced analytics and predictive insights, which can be incorporated in future iterations.

6. CONCLUSION AND FUTURE SCOPE

WasteChain works as a unified solution combining recycling and digital traceability. It makes waste management simpler, more transparent, and more structured. While not perfect yet, it provides a strong foundation for future development.

Future Scope

- AI-powered waste prediction and recommendations
- Mobile application development
- IoT-enabled smart waste bins
- Blockchain-based advanced traceability

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APPENDIX

Screenshots

Figure 1: Landing Page

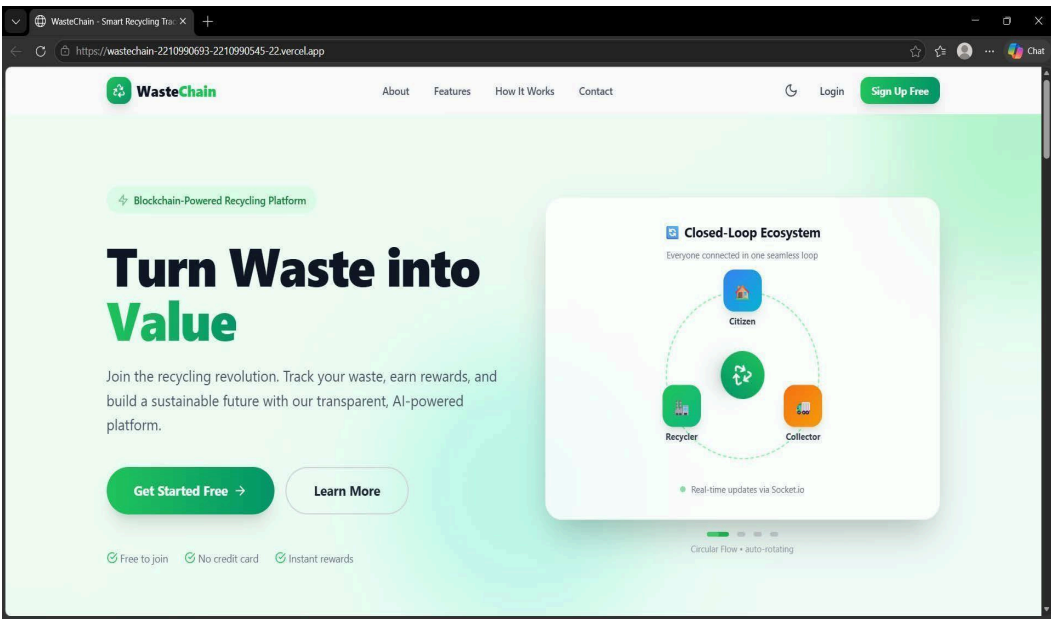


Figure 2: Dashboard

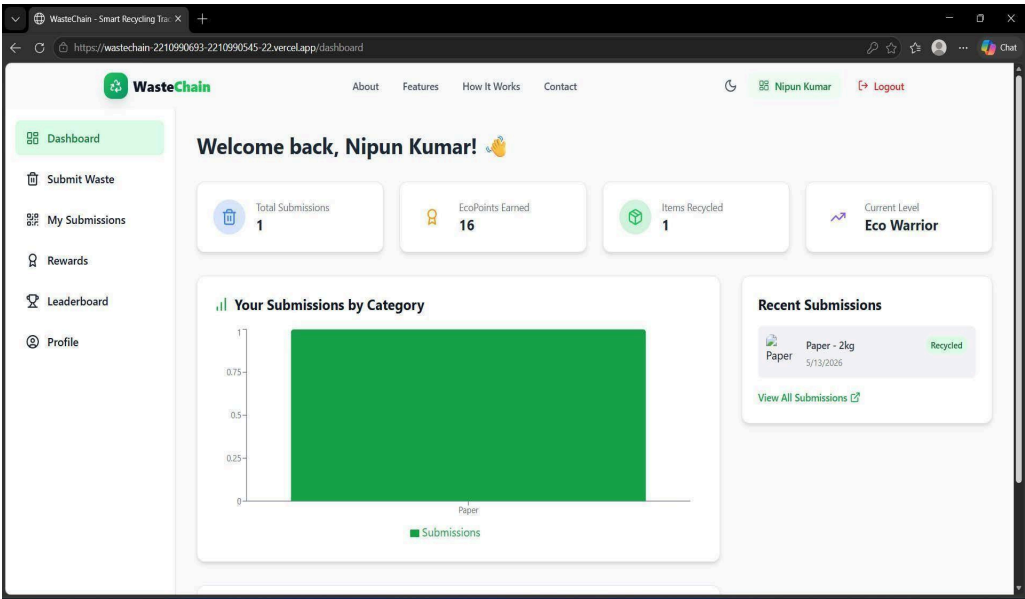


Figure 3: Submit Page

WasteChain

About Features How It Works Contact

Nipun Kumar Logout

Dashboard

Submit Waste

My Submissions

Rewards

Leaderboard

Profile

Submit Waste for Recycling

1. Upload Photo



Change Image

2. AI Analysis

Analyze Waste Type

3. Submission Details

Waste Category
Paper

Weight (kg, approx.)
2

Description
Old newspapers and waste paper collected for recycling

Pickup Address
Rajpura

Submit Waste

Figure 4: QR Page

WasteChain

About Features How It Works Contact

Nipun Kumar Logout

Track Your Waste



WC-1778654872056-NJYXS7QKN

Waste Tracking ID

Category	Weight
Paper	2 kg
Status	Points
Submitted	+16

Description
Old newspapers and waste paper collected for recycling

Submitted By
Nipun Kumar

Figure 5: Collector

